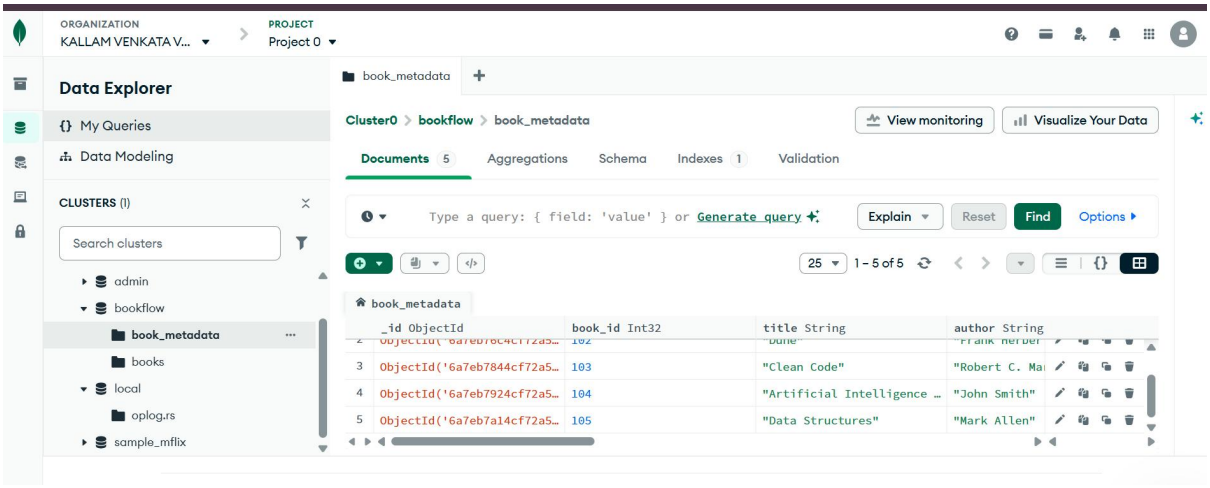
Database System Engineering and Distributed Backend Development

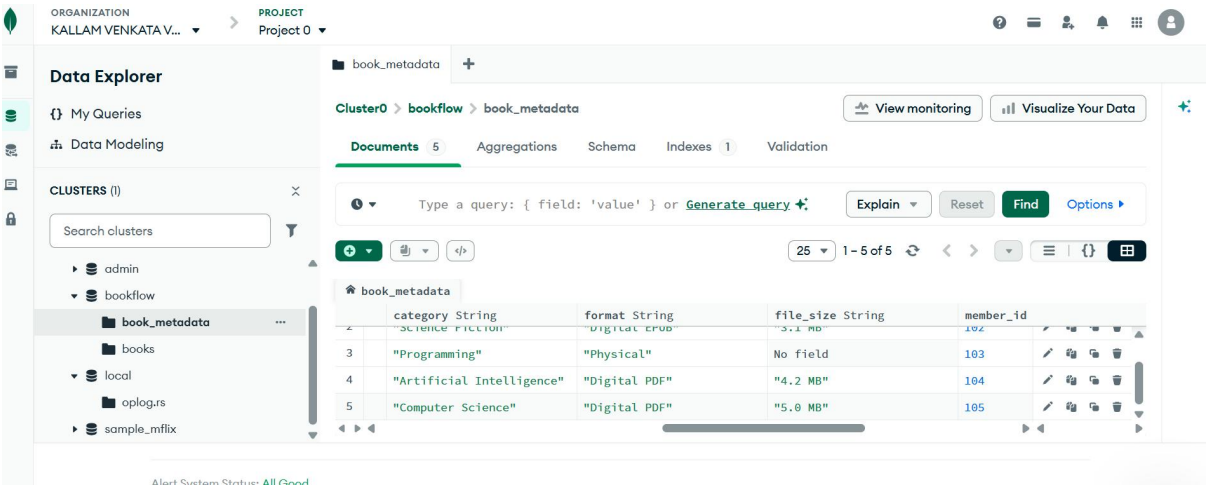
Course Code: 25CS13002E

Roll NO:2520030198

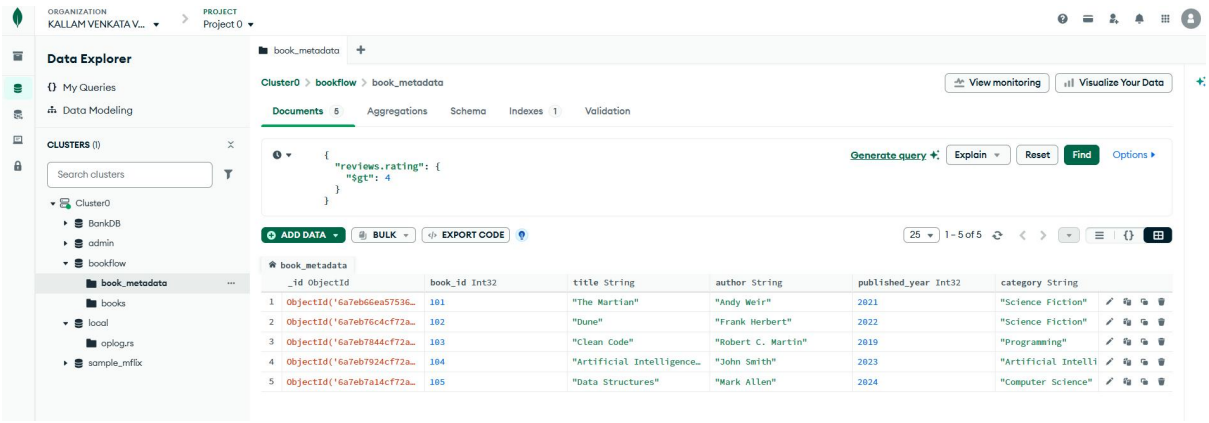
NAME: K.Vaishnavi

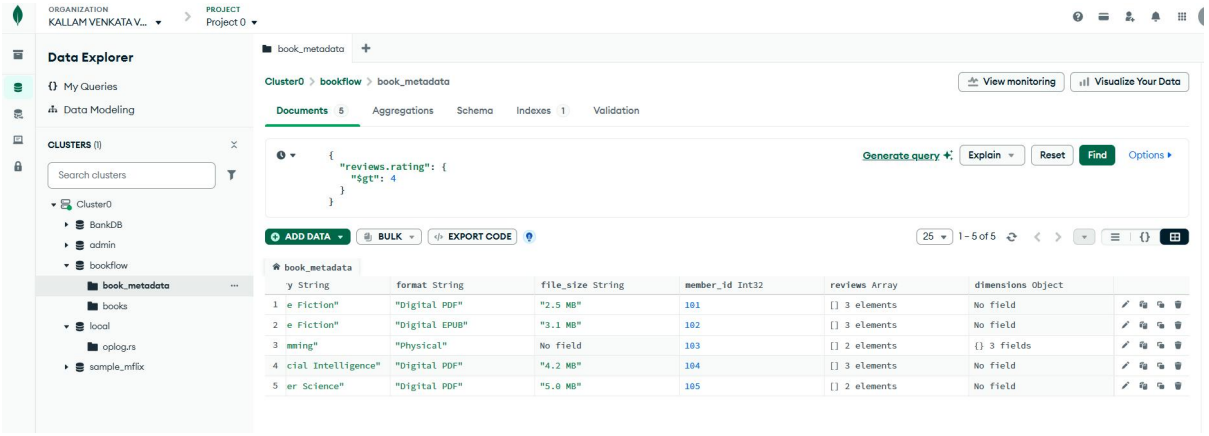
WEEK-3



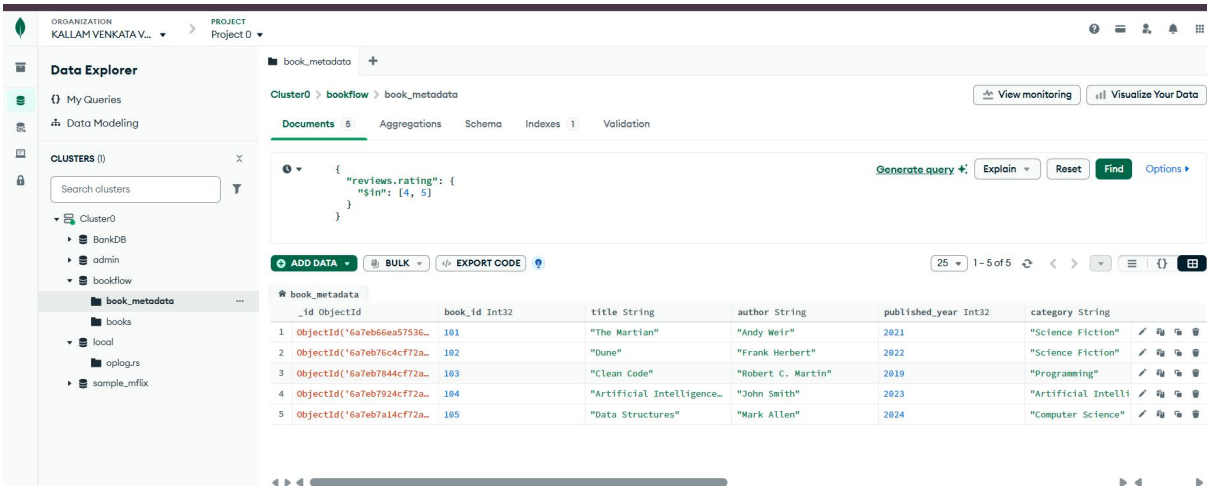


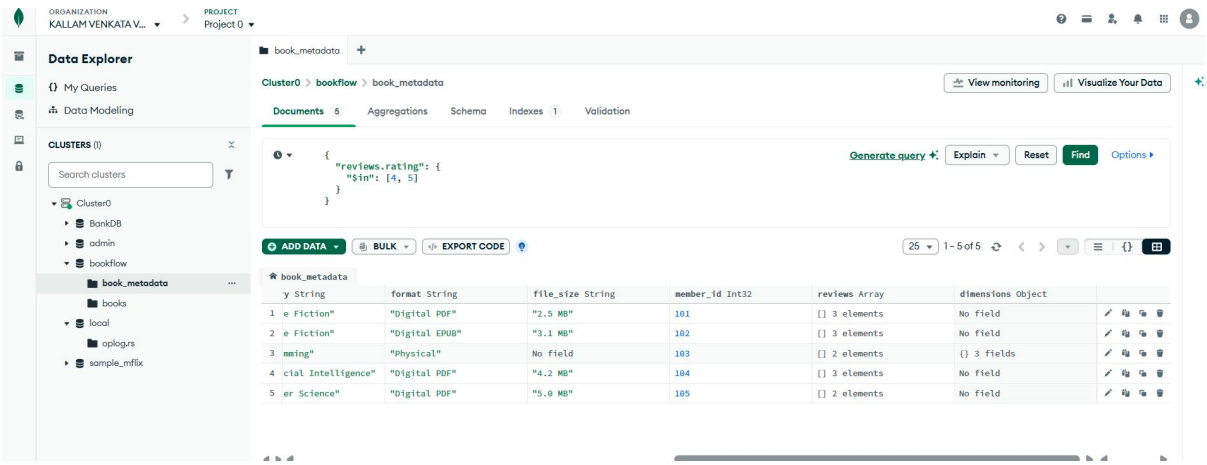
COMPARISON OPERATOR $gt


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Cluster0 > bookflow > book_metadata
View monitoring
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Indexes 1
Validation
{
"reviews.rating": {
"$gt": 4
}
}
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
25
1 - 5 of 5
book_metadata
_id ObjectId
book_id Int32
title String
author String
published_year Int32
category String
1 ObjectId('6a7eb66ea57536... 101 "The Martian" "Andy Weir" 2021 "Science Fiction"
2 ObjectId('6a7eb76c4cf72a... 102 "Dune" "Frank Herbert" 2022 "Science Fiction"
3 ObjectId('6a7eb7844cf72a... 103 "Clean Code" "Robert C. Martin" 2019 "Programming"
4 ObjectId('6a7eb7924cf72a... 104 "Artificial Intelligence... "John Smith" 2023 "Artificial Intelli
5 ObjectId('6a7eb7a14cf72a... 105 "Data Structures" "Mark Allen" 2024 "Computer Science"

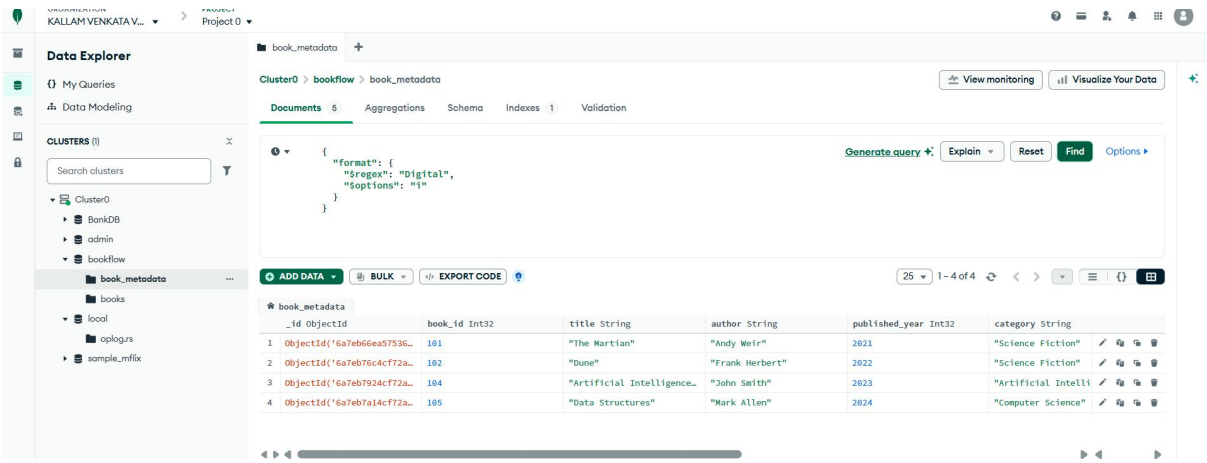

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Indexes 1
Validation
{
"reviews.rating": {
"$gt": 4
}
}
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
25
1 - 5 of 5
book_metadata
y String
format String
file_size String
member_id Int32
reviews Array
dimensions Object
1 e Fiction" "Digital PDF" "2.5 MB" 101 [] 3 elements No field
2 e Fiction" "Digital EPUB" "3.1 MB" 102 [] 3 elements No field
3 mming" "Physical" No field 103 [] 2 elements {} 3 fields
4 cial Intelligence" "Digital PDF" "4.2 MB" 104 [] 3 elements No field
5 er Science" "Digital PDF" "5.0 MB" 105 [] 2 elements No field

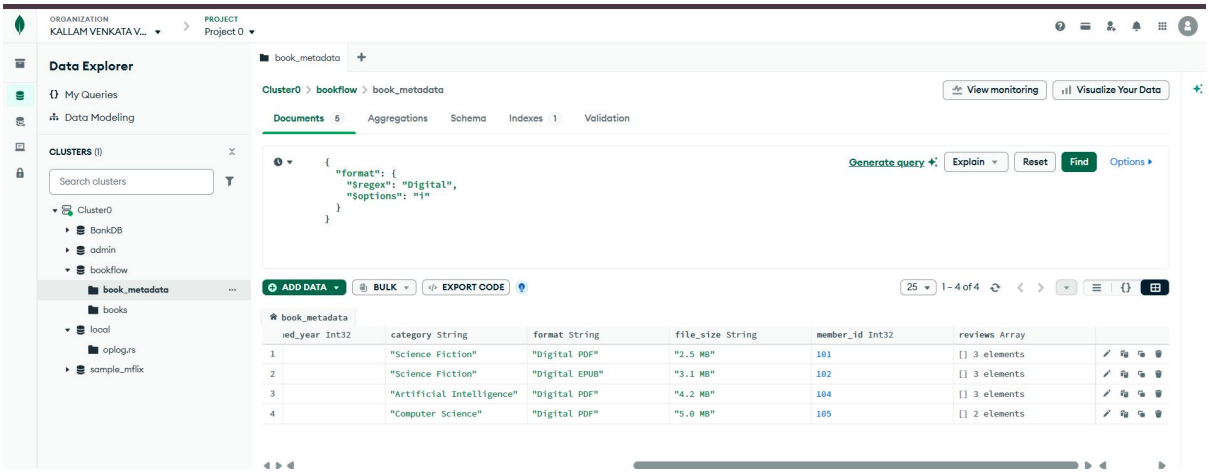
USING $IN


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book_metadata
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Visualize Your Data
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Aggregations
Schema
Indexes 1
Validation
{
"reviews.rating": {
"$in": [4, 5]
}
}
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
25
1 - 5 of 5
book_metadata
_id ObjectId
book_id Int32
title String
author String
published_year Int32
category String
1 ObjectId('6a7eb66ea57536... 101 "The Martian" "Andy Weir" 2021 "Science Fiction"
2 ObjectId('6a7eb76c4cf72a... 102 "Dune" "Frank Herbert" 2022 "Science Fiction"
3 ObjectId('6a7eb7844cf72a... 103 "Clean Code" "Robert C. Martin" 2019 "Programming"
4 ObjectId('6a7eb7924cf72a... 104 "Artificial Intelligence... "John Smith" 2023 "Artificial Intelli
5 ObjectId('6a7eb7a14cf72a... 105 "Data Structures" "Mark Allen" 2024 "Computer Science"

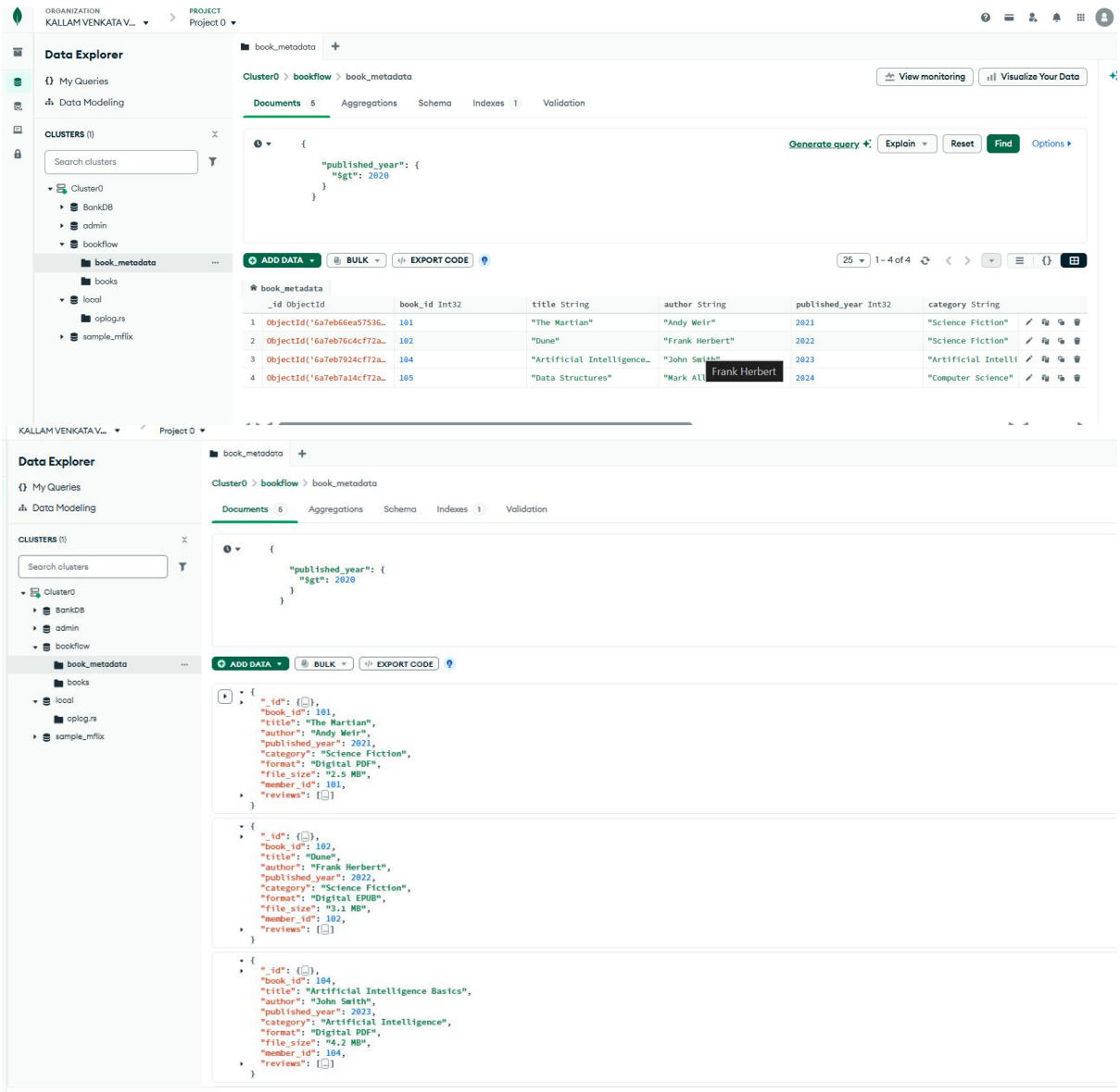

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Validation
{
"reviews.rating": {
"$in": [4, 5]
}
}
Generate query
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25
1 - 5 of 5
book_metadata
y String
format String
file_size String
member_id Int32
reviews Array
dimensions Object
1 e Fiction" "Digital PDF" "2.5 MB" 101 [] 3 elements No field
2 e Fiction" "Digital EPUB" "3.1 MB" 102 [] 3 elements No field
3 mming" "Physical" No field 103 [] 2 elements {} 3 fields
4 cial Intelligence" "Digital PDF" "4.2 MB" 104 [] 3 elements No field
5 er Science" "Digital PDF" "5.0 MB" 105 [] 2 elements No field

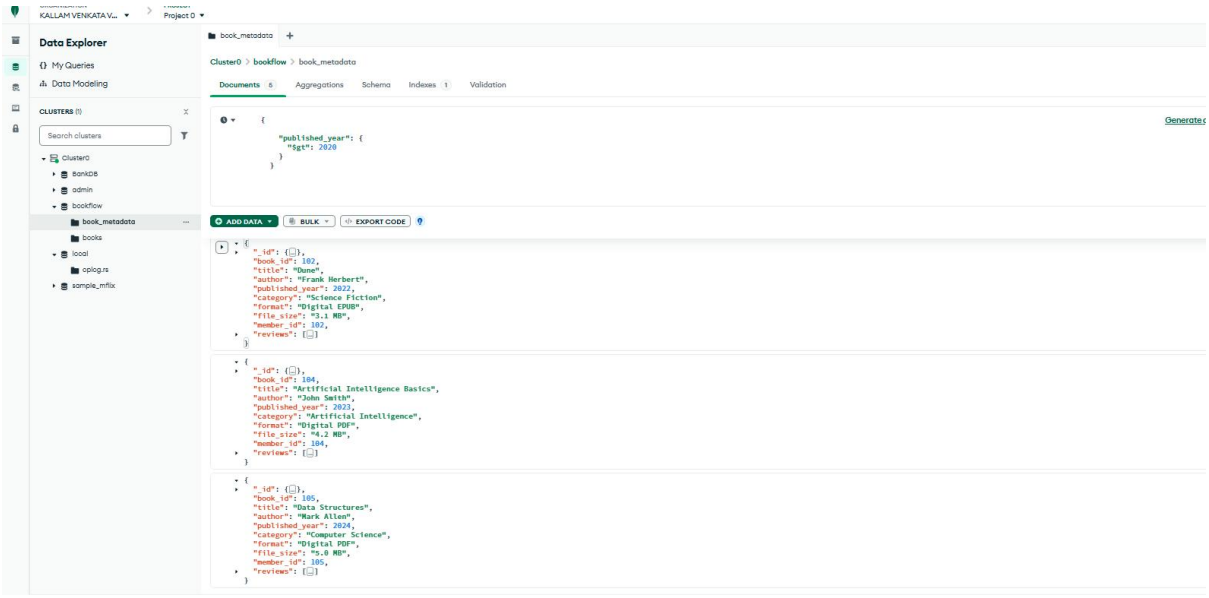
REGEX QUERY


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Indexes 1
Validation
{
"format": {
"$regex": "Digital",
"$options": "i"
}
}
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
25
1 - 4 of 4
book_metadata
_id ObjectId
book_id Int32
title String
author String
published_year Int32
category String
1 ObjectId('6a7eb66ea57536... 101 "The Martian" "Andy Weir" 2021 "Science Fiction"
2 ObjectId('6a7eb76c4cf72a... 102 "Dune" "Frank Herbert" 2022 "Science Fiction"
3 ObjectId('6a7eb7924cf72a... 104 "Artificial Intelligence... "John Smith" 2023 "Artificial Intelli
4 ObjectId('6a7eb7a14cf72a... 105 "Data Structures" "Mark Allen" 2024 "Computer Science"

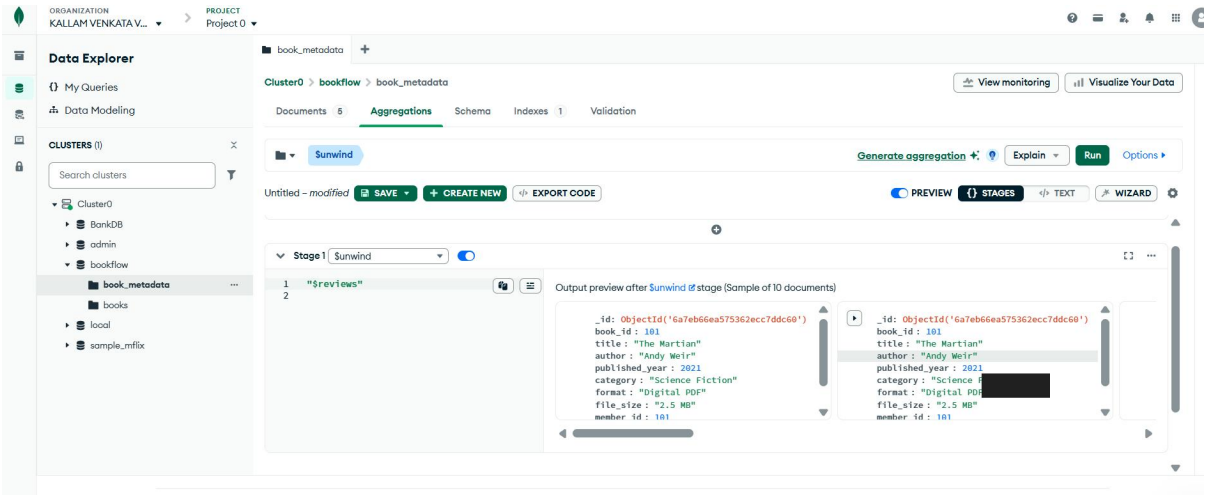

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Indexes 1
Validation
{
"format": {
"$regex": "Digital",
"$options": "i"
}
}
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
25
1 - 4 of 4
book_metadata
ed_year Int32
category String
format String
file_size String
member_id Int32
reviews Array
1 "Science Fiction" "Digital PDF" "2.5 MB" 101 [] 3 elements
2 "Science Fiction" "Digital EPUB" "3.1 MB" 102 [] 3 elements
3 "Artificial Intelligence" "Digital PDF" "4.2 MB" 104 [] 3 elements
4 "Computer Science" "Digital PDF" "5.0 MB" 105 [] 2 elements

$MATCH

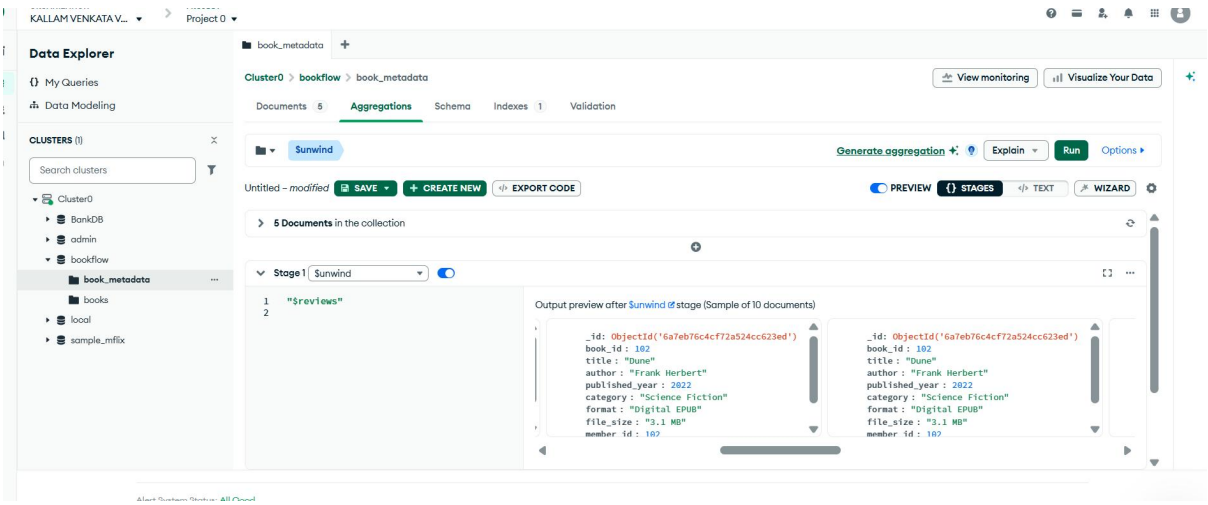

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Cluster0 > bookflow > book_metadata
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Schema
Indexes 1
Validation
{
"published_year": {
"$gt": 2020
}
}
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
25
1 - 4 of 4
book_metadata
_id ObjectId
book_id Int32
title String
author String
published_year Int32
category String
1 ObjectId('6a7eb66ea57536... 101 "The Martian" "Andy Weir" 2021 "Science Fiction"
2 ObjectId('6a7eb76c4cf72a... 102 "Dune" "Frank Herbert" 2022 "Science Fiction"
3 ObjectId('6a7eb7924cf72a... 104 "Artificial Intelligence... "John Smith" 2023 "Artificial Intelli
4 ObjectId('6a7eb7a14cf72a... 105 "Data Structures" "Mark Al 2024 "Computer Science"
Frank Herbert
KALLAM VENKATA V...
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Cluster0 > bookflow > book_metadata
Documents 6
Aggregations
Schema
Indexes 1
Validation
{
"published_year": {
"$gt": 2020
}
}
ADD DATA
BULK
EXPORT CODE
{
"_id": {},
"book_id": 101,
"title": "The Martian",
"author": "Andy Weir",
"published_year": 2021,
"category": "Science Fiction",
"format": "Digital PDF",
"file_size": "2.5 MB",
"member_id": 101,
"reviews": []
}
{
"_id": {},
"book_id": 102,
"title": "Dune",
"author": "Frank Herbert",
"published_year": 2022,
"category": "Science Fiction",
"format": "Digital EPUB",
"file_size": "3.1 MB",
"member_id": 102,
"reviews": []
}
{
"_id": {},
"book_id": 104,
"title": "Artificial Intelligence Basics",
"author": "John Smith",
"published_year": 2023,
"category": "Artificial Intelligence",
"format": "Digital PDF",
"file_size": "4.2 MB",
"member_id": 104,
"reviews": []
}

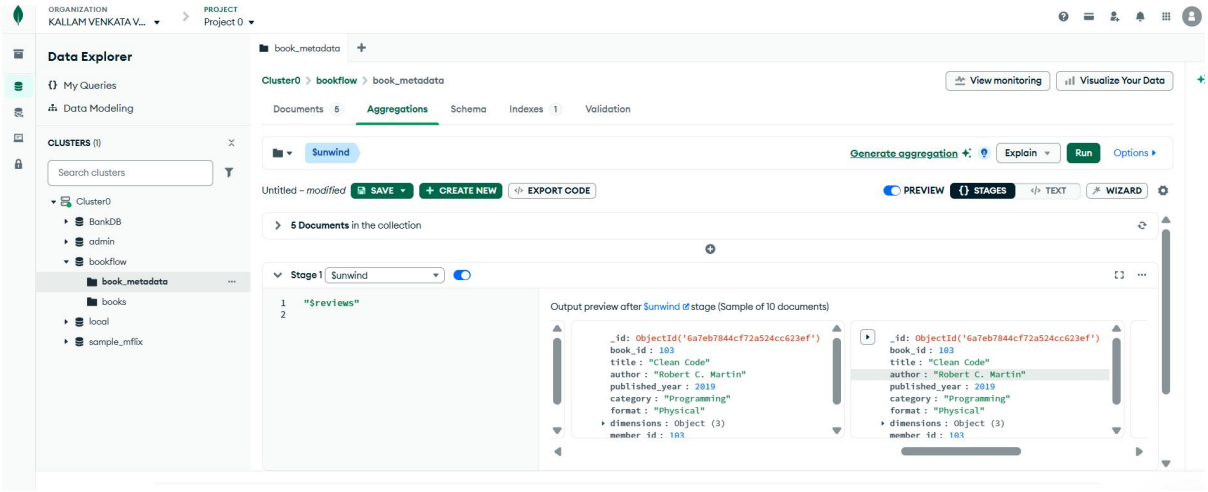


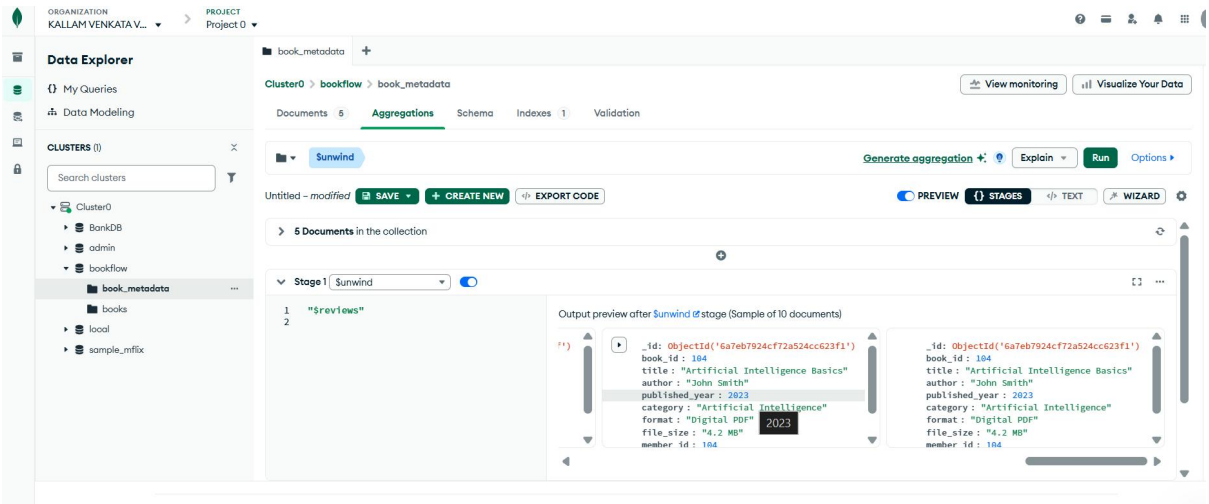
$UNWIND



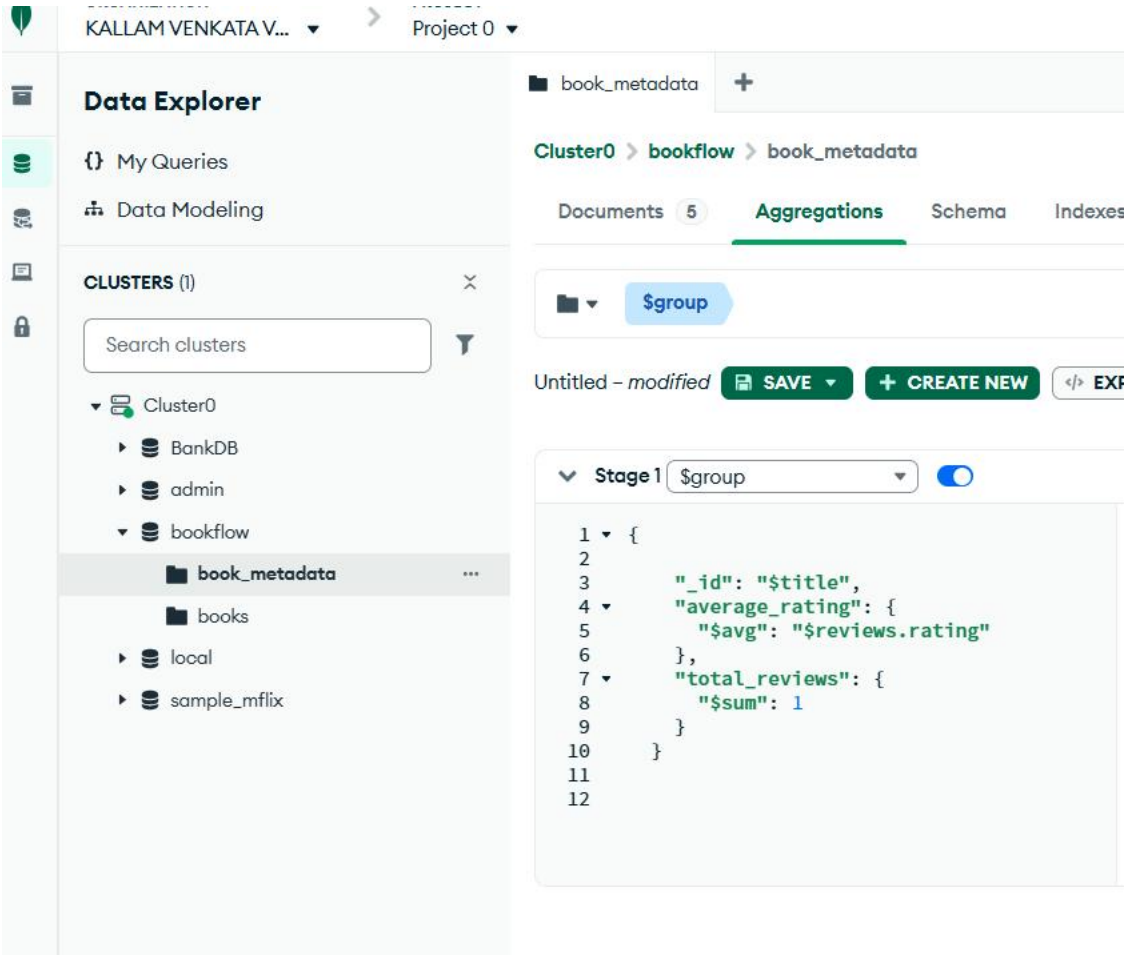
Output: -


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Options
Untitled – modified
SAVE
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EXPORT CODE
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STAGES
TEXT
WIZARD
5 Documents in the collection
Stage 1
Sunwind
"$reviews"
Output preview after $unwind stage (Sample of 10 documents)
_id: ObjectId('6a7eb76c4cf72a524cc623ed')
book_id : 102
title : "Dune"
author : "Frank Herbert"
published_year : 2022
category : "Science Fiction"
format : "Digital EPUB"
file_size : "3.1 MB"
member_id : 102

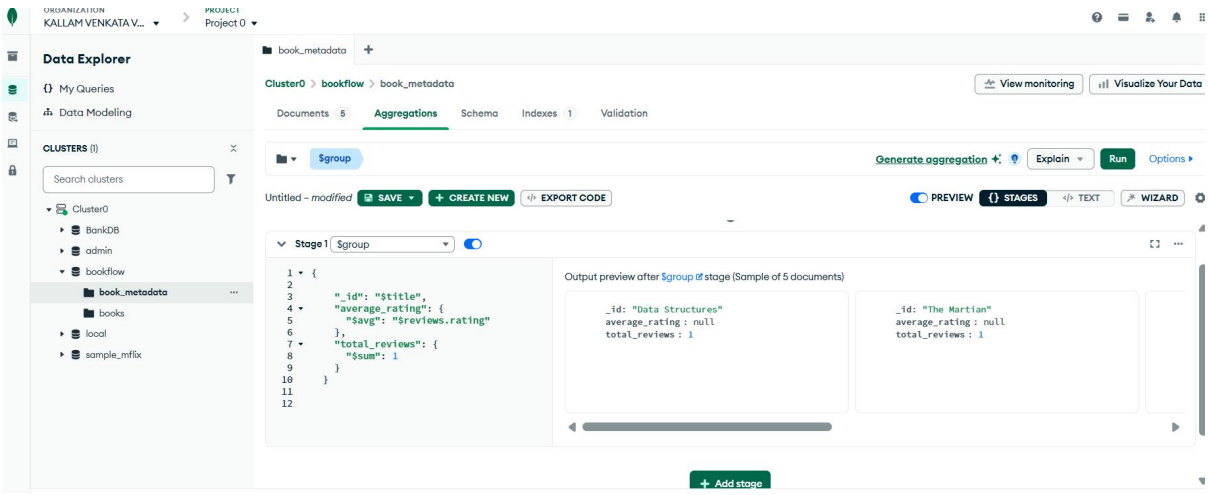

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Untitled – modified
SAVE
CREATE NEW
EXPORT CODE
PREVIEW
STAGES
TEXT
WIZARD
5 Documents in the collection
Stage 1
Sunwind
"$reviews"
Output preview after $unwind stage (Sample of 10 documents)
_id: ObjectId('6a7eb7844cf72a524cc623ef')
book_id : 103
title : "Clean Code"
author : "Robert C. Martin"
published_year : 2019
category : "Programming"
format : "Physical"
dimensions : Object (3)
member_id : 103

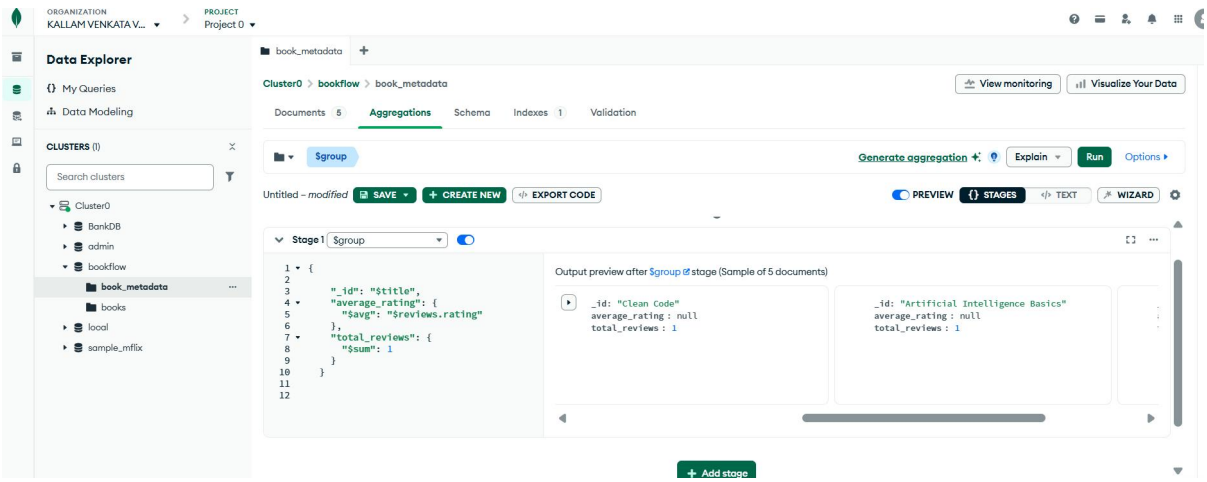

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Options
Untitled – modified
SAVE
CREATE NEW
EXPORT CODE
PREVIEW
STAGES
TEXT
WIZARD
5 Documents in the collection
Stage 1
Sunwind
"$reviews"
Output preview after $unwind stage (Sample of 10 documents)
_id: ObjectId('6a7eb7924cf72a524cc623f1')
book_id : 104
title : "Artificial Intelligence Basics"
author : "John Smith"
published_year : 2023
category : "Artificial Intelligence"
format : "Digital PDF"
file_size : "4.2 MB"
member_id : 104
2023

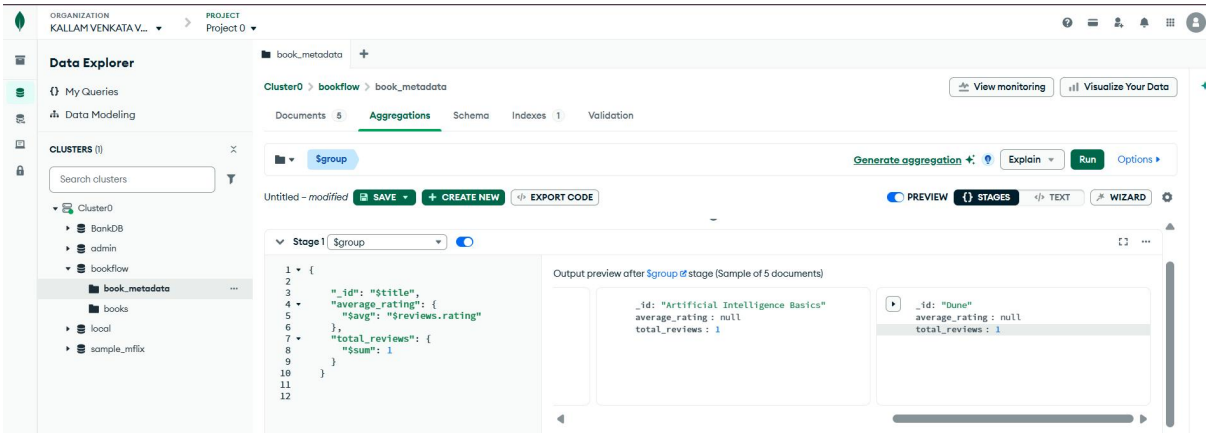
$GROUP


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$group
Untitled – modified
SAVE
+ CREATE NEW
Stage 1
$group
{
"_id": "$title",
"average_rating": {
"$avg": "$reviews.rating"
},
"total_reviews": {
"$sum": 1
}
}

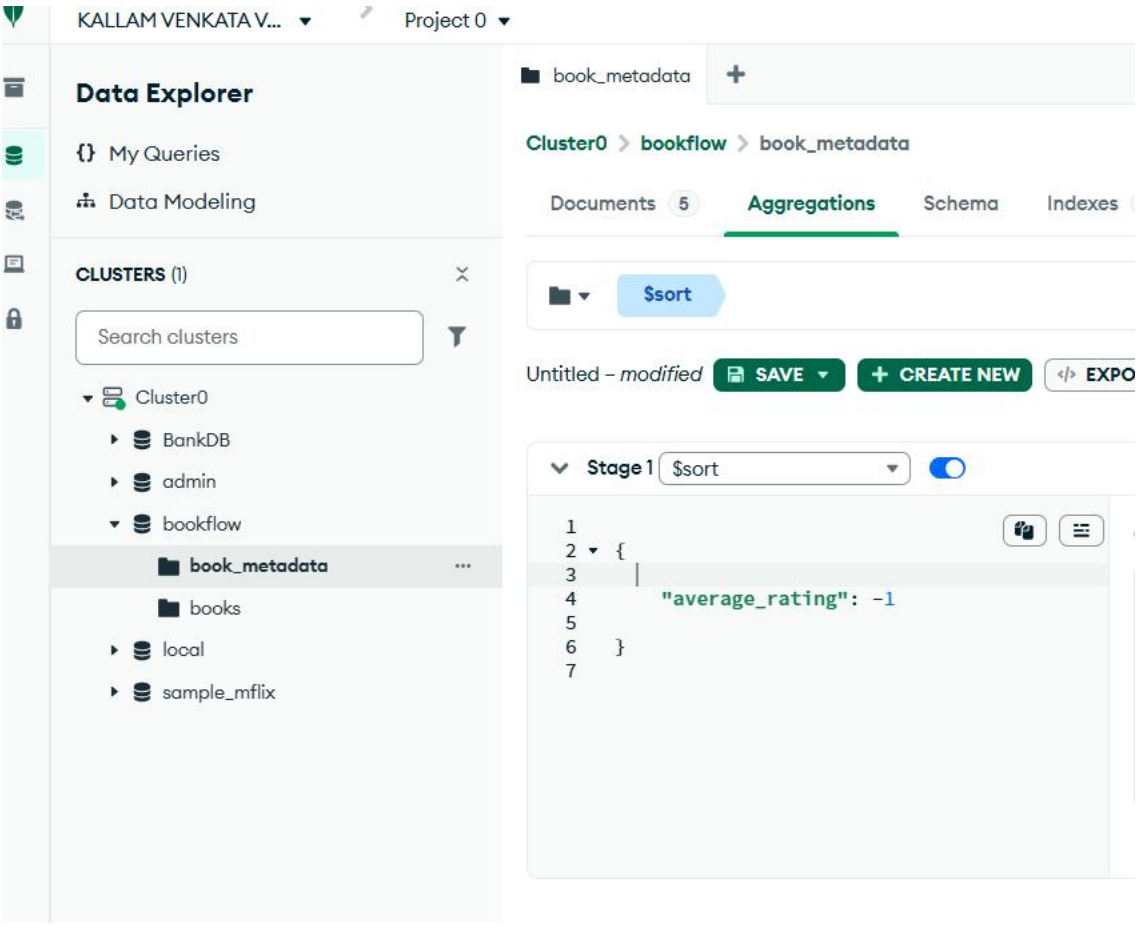
Output: -


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Validation
$group
Generate aggregation
Explain
Run
Options
Untitled – modified
SAVE
+ CREATE NEW
EXPORT CODE
PREVIEW
STAGES
TEXT
WIZARD
Stage 1
$group
Output preview after $group stage (Sample of 5 documents)
_id: "Data Structures"
average_rating : null
total_reviews : 1
_id: "The Martian"
average_rating : null
total_reviews : 1
+ Add stage

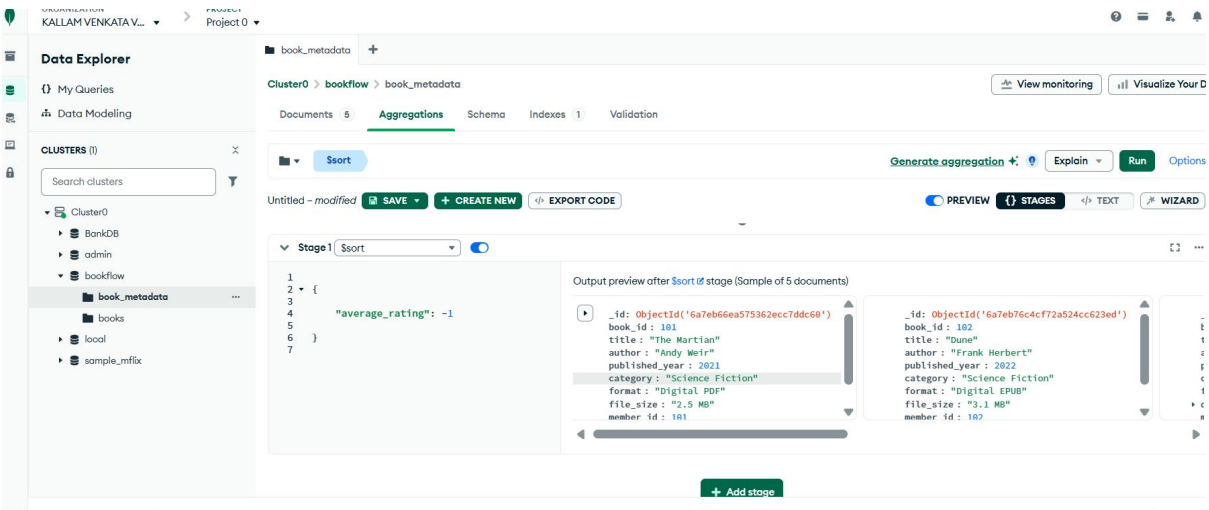

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Validation
$group
Generate aggregation
Explain
Run
Options
Untitled – modified
SAVE
CREATE NEW
EXPORT CODE
PREVIEW
STAGES
TEXT
WIZARD
Stage 1 $group
{
"_id": "$title",
"average_rating": {
"$avg": "$reviews.rating"
},
"total_reviews": {
"$sum": 1
}
}
Output preview after $group stage (Sample of 5 documents)
_id: "Clean Code"
average_rating : null
total_reviews : 1
_id: "Artificial Intelligence Basics"
average_rating : null
total_reviews : 1
Add stage

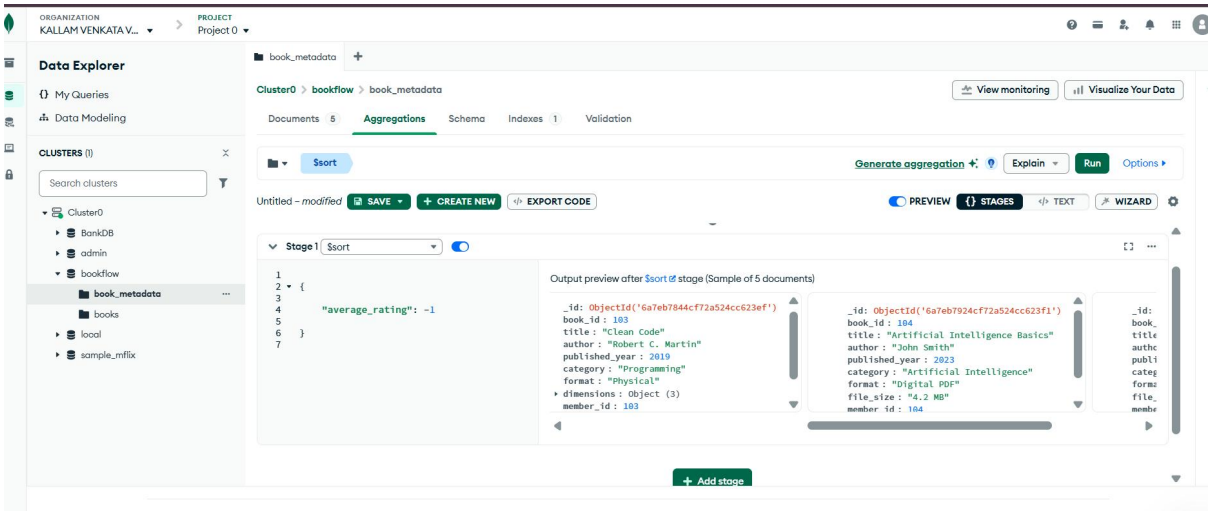

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$group
Generate aggregation
Explain
Run
Options
Untitled – modified
SAVE
CREATE NEW
EXPORT CODE
PREVIEW
STAGES
TEXT
WIZARD
Stage 1 $group
{
"_id": "$title",
"average_rating": {
"$avg": "$reviews.rating"
},
"total_reviews": {
"$sum": 1
}
}
Output preview after $group stage (Sample of 5 documents)
_id: "Artificial Intelligence Basics"
average_rating : null
total_reviews : 1
_id: "Dune"
average_rating : null
total_reviews : 1

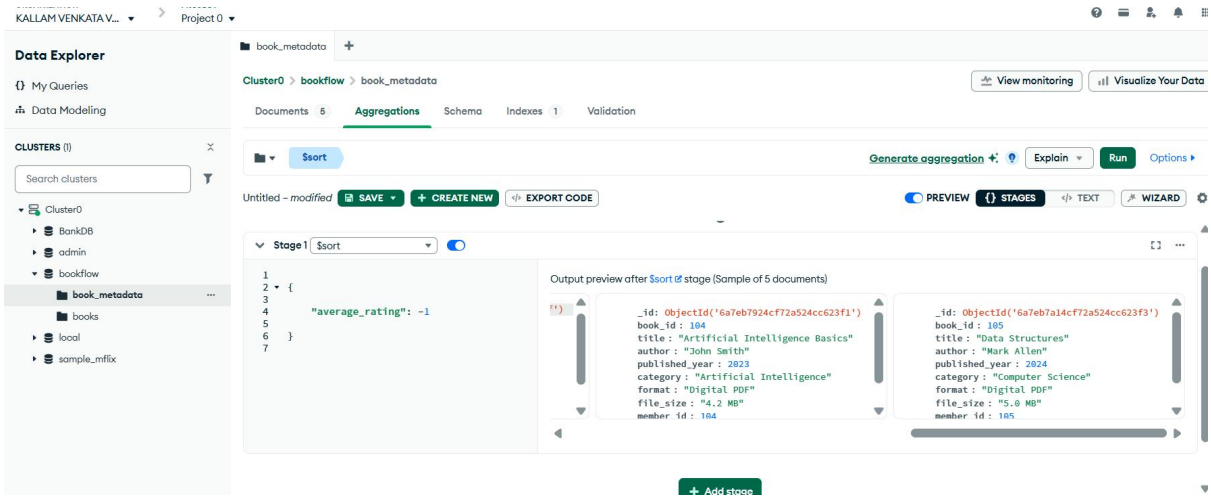
$SORT


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Stage 1
$sort
{
"average_rating": -1
}

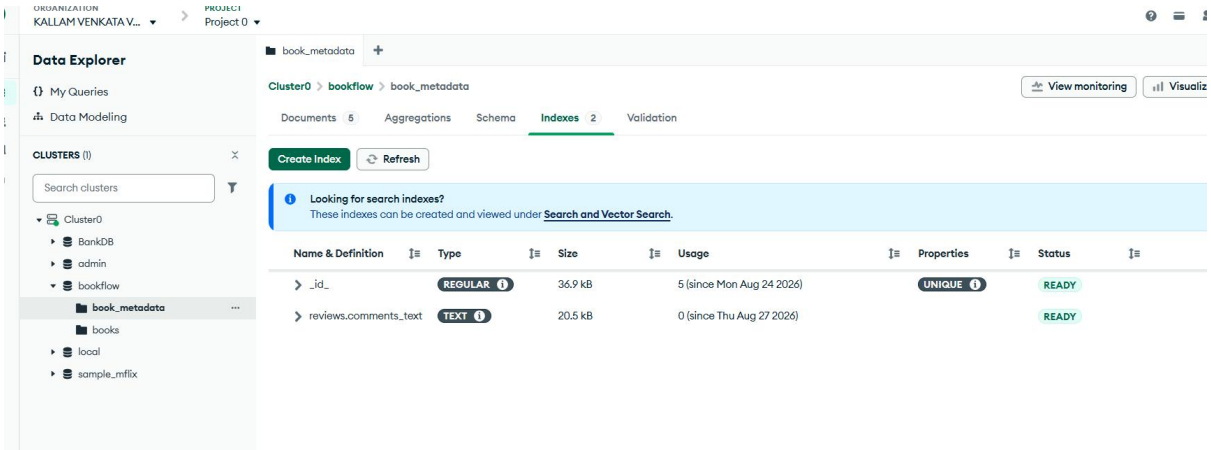
Output: -


Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
BankDB
admin
bookflow
book_metadata
books
local
sample_mflix
book_metadata
Cluster0 > bookflow > book_metadata
View monitoring
Documents 5
Aggregations
Schema
Indexes 1
Validation
$sort
Generate aggregation
Explain
Run
Options
Untitled – modified
SAVE
CREATE NEW
EXPORT CODE
PREVIEW
STAGES
TEXT
WIZARD
Stage 1
$sort
{
"average_rating": -1
}
Output preview after $sort stage (Sample of 5 documents)
_id: ObjectId('6a7eb66ea575362ecc7ddc60')
book_id : 101
title : "The Martian"
author : "Andy Weir"
published_year : 2021
category : "Science Fiction"
format : "Digital PDF"
file_size : "2.5 MB"
_id: ObjectId('6a7eb76c4cf72a524cc623ed')
book_id : 102
title : "Dune"
author : "Frank Herbert"
published_year : 2022
category : "Science Fiction"
format : "Digital EPUB"
file_size : "3.1 MB"
Add stage

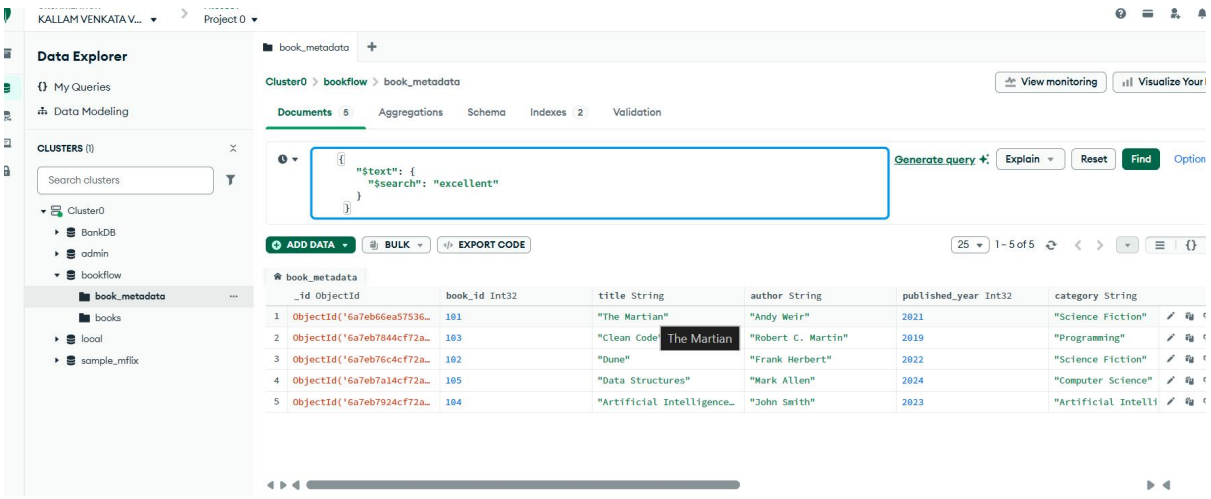


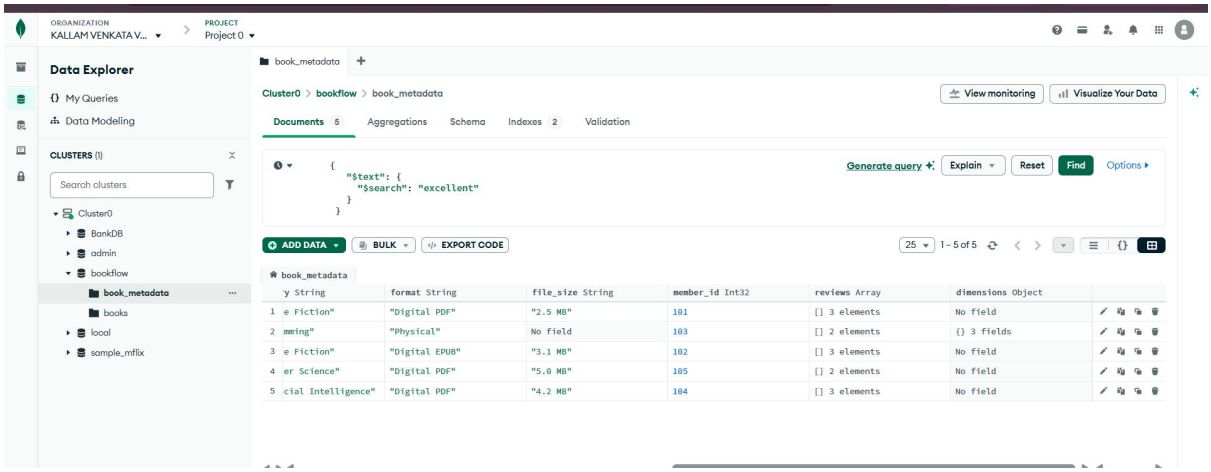


CREATE TEXT INDEX

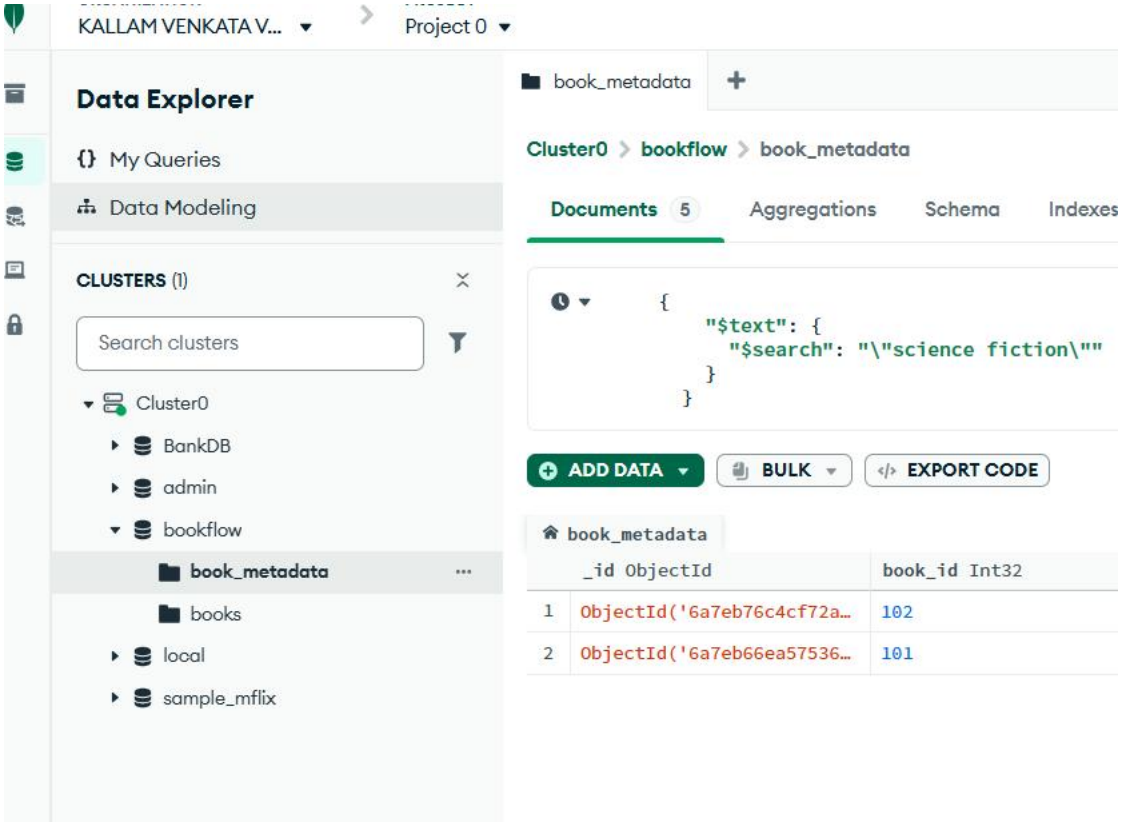


SEARCH USING TEXT INDEX

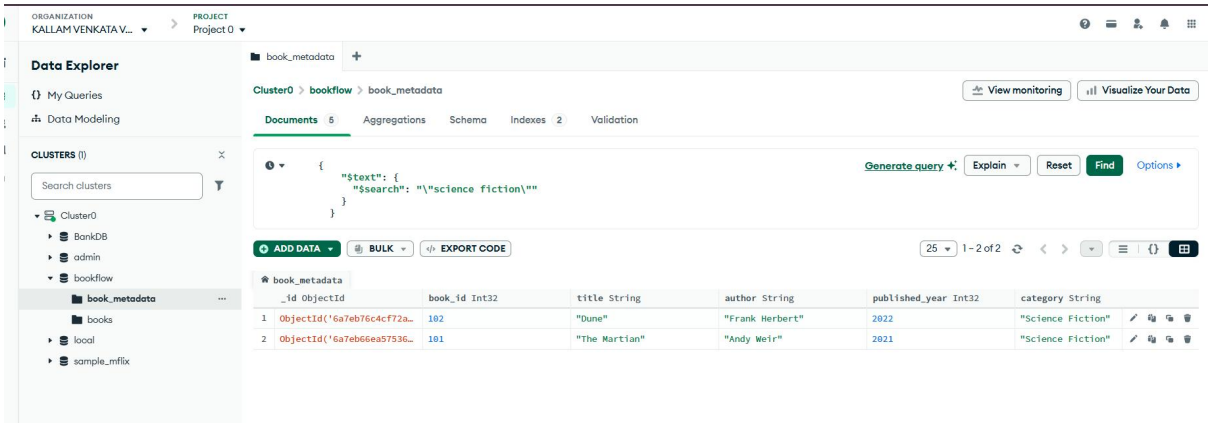


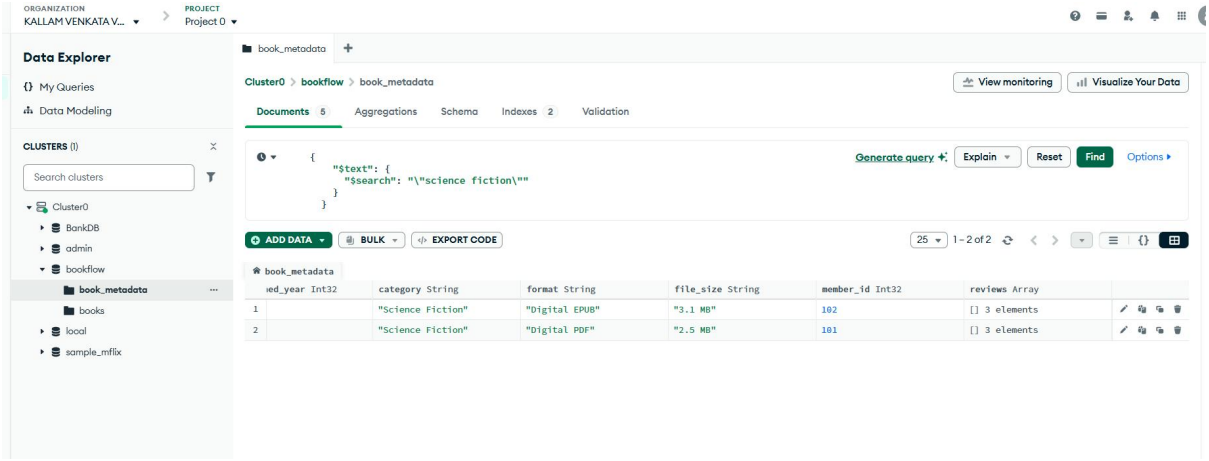


SEARCH FOR A SPECIFIC KEYWORD

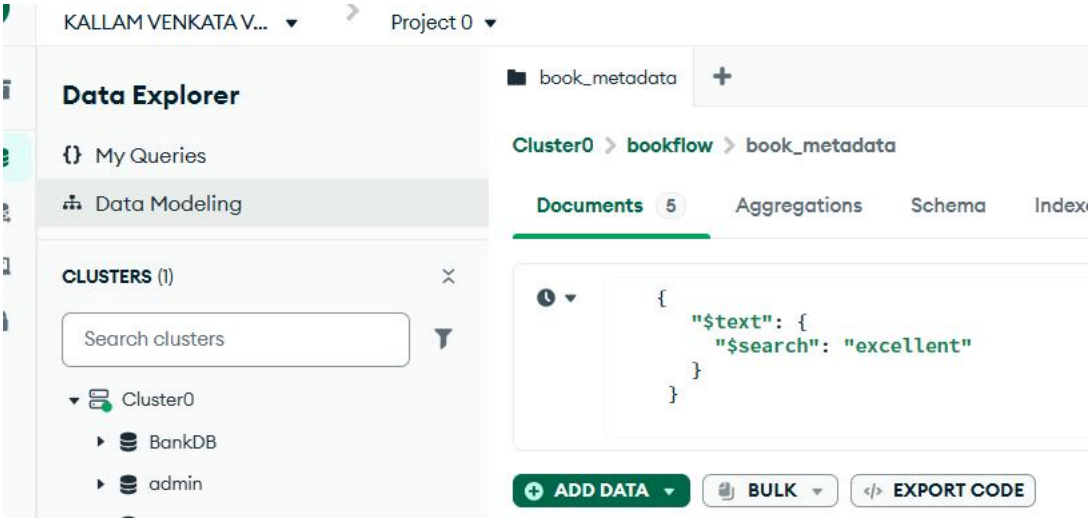

KALLAM VENKATA V...
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
BankDB
admin
bookflow
book_metadata
books
local
sample_mflix
book_metadata
Cluster0 > bookflow > book_metadata
Documents 5
Aggregations
Schema
Indexes
{
"$text": {
"$search": "\"science fiction\""
}
}
ADD DATA
BULK
EXPORT CODE
book_metadata
_id ObjectId
book_id Int32
1
ObjectId('6a7eb76c4cf72a...
102
2
ObjectId('6a7eb66ea57536...
101

Output: -


ORGANIZATION
KALLAM VENKATA V...
PROJECT
Project 0
Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
Cluster0
BankDB
admin
bookflow
book_metadata
books
local
sample_mflix
book_metadata
Cluster0 > bookflow > book_metadata
View monitoring
Visualize Your Data
Documents 5
Aggregations
Schema
Indexes 2
Validation
{
"$text": {
"$search": "\"science fiction\""
}
}
Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT CODE
25
1 - 2 of 2
book_metadata
_id ObjectId
book_id Int32
title String
author String
published_year Int32
category String
1
ObjectId('6a7eb76c4cf72a...
102
"Dune"
"Frank Herbert"
2022
"Science Fiction"
2
ObjectId('6a7eb66ea57536...
101
"The Martian"
"Andy Weir"
2021
"Science Fiction"



USE explain()



Output: -

